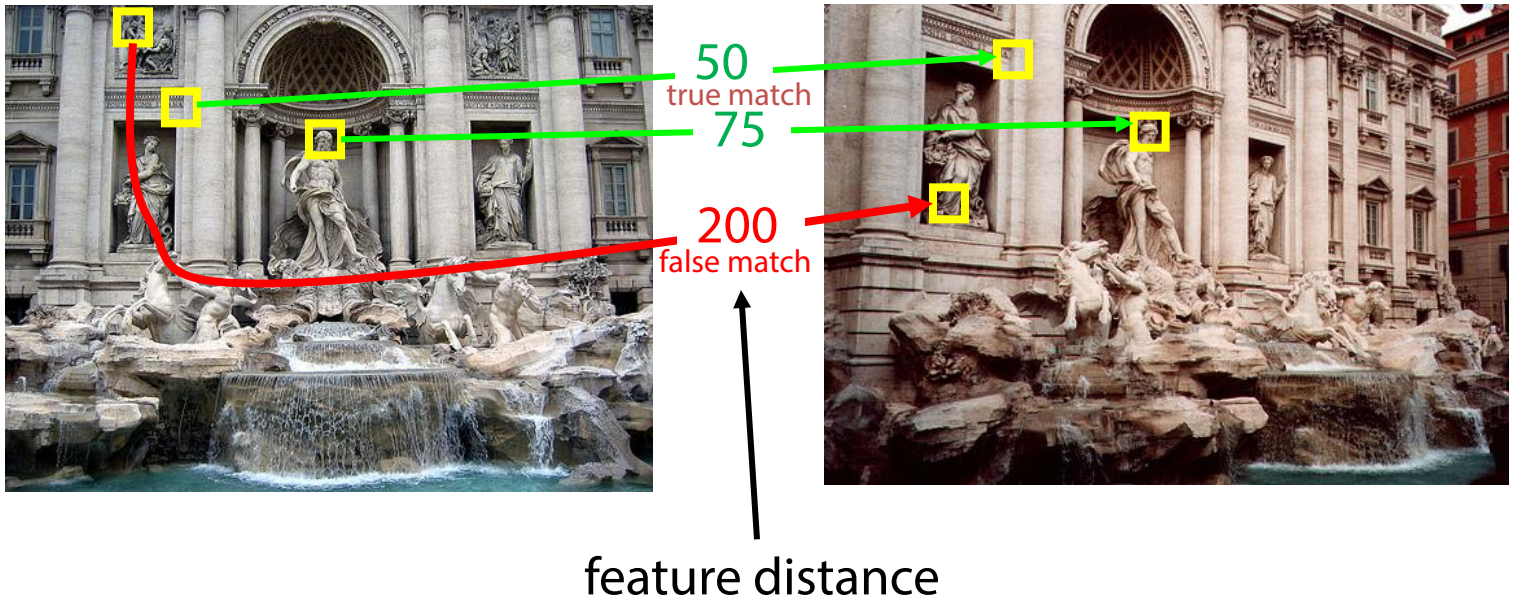
True/false positives

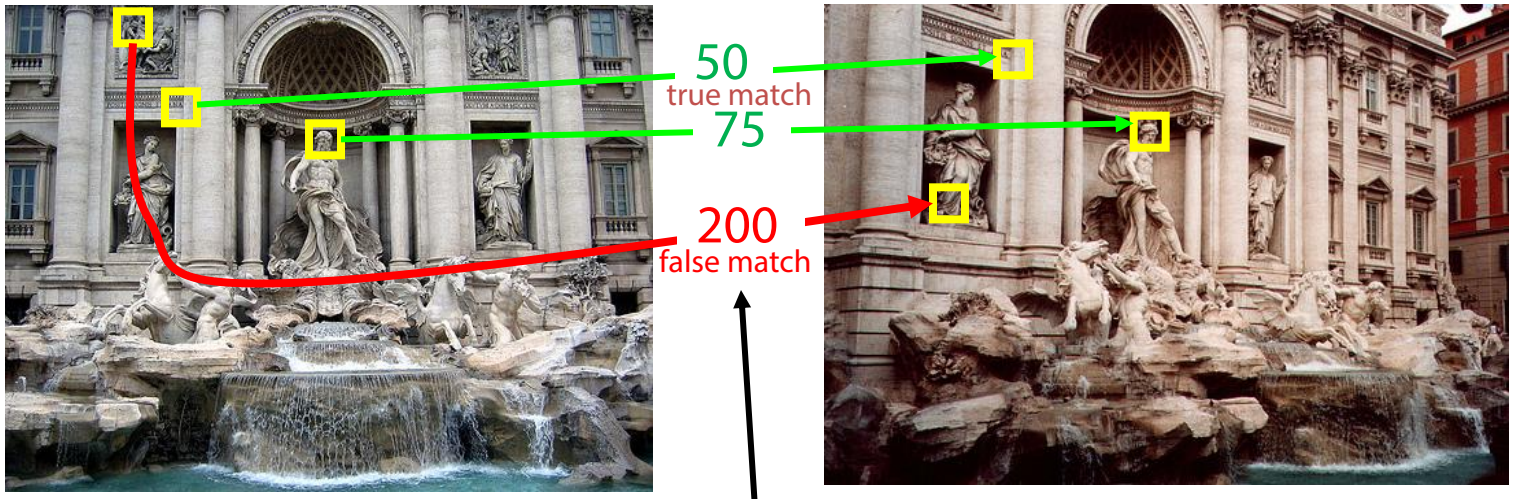
How can we measure the performance of a feature matcher?



Suppose we want to **maximize true positives**. How do we set the threshold?
(Note: we keep all matches with distance below the threshold.)

True/false positives

How can we measure the performance of a feature matcher?



Suppose we want to **minimize false positives**. How do we set the threshold?
(Note: we keep all matches with distance below the threshold.)

Example

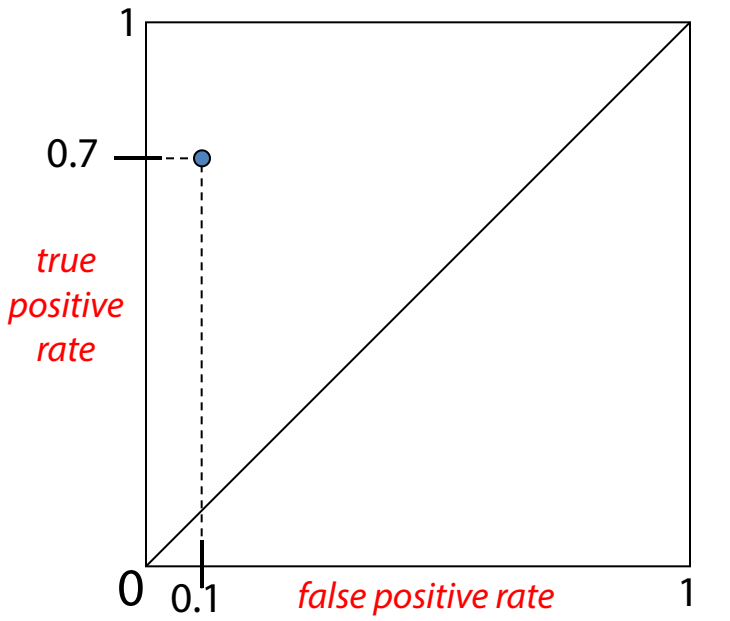
- Suppose our matcher computes 1,000 matches between two images
  - 800 are correct matches, 200 are incorrect (according to an oracle that gives us ground truth matches)
  - A given threshold (e.g., ratio distance = 0.6) gives us 600 correct matches and 100 incorrect matches that survive the threshold
  - True positive rate = 600 / 800 = ¾
  - False positive rate = 100 / 200 = ½

Evaluating the results

How can we measure the performance of a feature matcher?

$$\frac{\text{\# true positives surviving threshold}}{\text{\# total correct matches (positives)}}$$

recall



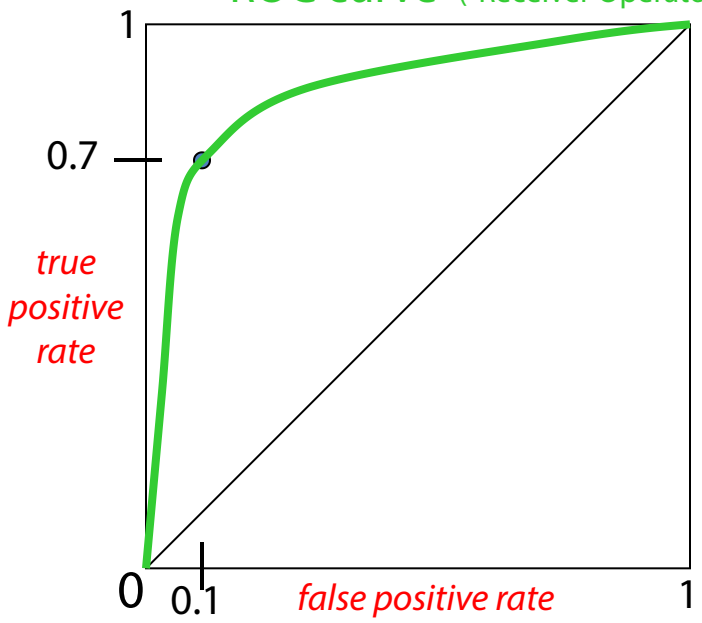
$$\frac{\text{\# false positives surviving threshold}}{\text{\# total incorrect matches (negatives)}}$$

1 - *specificity*

Evaluating the results

How can we measure the performance of a feature matcher?

ROC curve ("Receiver Operator Characteristic")



$$\frac{\text{\# true positives surviving threshold}}{\text{\# total correct matches (positives)}}$$

recall

$$\frac{\text{\# false positives surviving threshold}}{\text{\# total incorrect matches (negatives)}}$$

1 - *specificity*

Single number: Area Under the Curve (AUC)

E.g. AUC = 0.87

1 is the best

ROC curves – summary

- By thresholding the match distances at different thresholds, we can generate sets of matches with different true/false positive rates
- ROC curve is generated by computing rates at a set of threshold values swept through the full range of possible thresholds
- Area under the ROC curve (AUC) summarizes the performance of a feature pipeline (higher AUC is better)

Lots of applications

Features are used for:

- Image alignment (e.g., mosaics)
- 3D reconstruction
- Motion tracking
- Object recognition
- Indexing and database retrieval
- Robot navigation
- … other

Object recognition (David Lowe)



